# 1 Preliminary Experiment

To validate the core game dynamics and verify that curriculum training can induce a strategically coherent equilibrium, we conducted a preliminary experiment using a lighter architecture—a multi-layer perceptron (MLP) with REINFORCE—on a version of OracleGambit without the balance system. Training proceeded in three phases (Fig. 1).

**Phase A – Honest Host, Players Only.** The host was fixed as a perfectly honest oracle (Honesty = 1.0) and only the players were trained. Across 18,048 rounds, the follow rate rose from 0.25 (random chance) to 0.69, with the win rate matching it exactly—confirming that players successfully learned to exploit a truthful signal.

**Phase B – Deceptive Host, Host Only.** Players were frozen and only the host was trained to maximise its own reward by deceiving players. Over rounds 20,096–42,112, signal honesty collapsed from 0.24 to 0.04, while the host's win rate dropped from 0.24 to 0.11. This seemingly counterintuitive result arises because the frozen players continued to follow a signal that was now systematically false, driving both parties toward anti-coordinated outcomes.

**Phase C – Joint Training (500 checkpoints).** Both agents were unfrozen and trained simultaneously for over one million rounds (44,160–1,042,176). As shown in Fig. 2, an *adversarial phase* dominated early Phase C: the host briefly exploited residual player trust (follow rate $\approx 0.56$ at the phase start) before players adapted. The gap $\Delta =$ Follow Rate $-$ Signal Honesty oscillated around zero and steadily contracted (Fig. 3), ultimately converging to the theoretical mixed-strategy Nash equilibrium at Honesty $\approx$ Follow Rate $\approx 0.25$—equal to the random-chance baseline of $1/D$ ($D = 4$ doors).

These results confirm that (i) the zero-sum payout formula is correctly implemented, (ii) curriculum shaping is necessary to avoid trivial convergence, and (iii) the system reliably finds the mixed-strategy equilibrium under joint training.

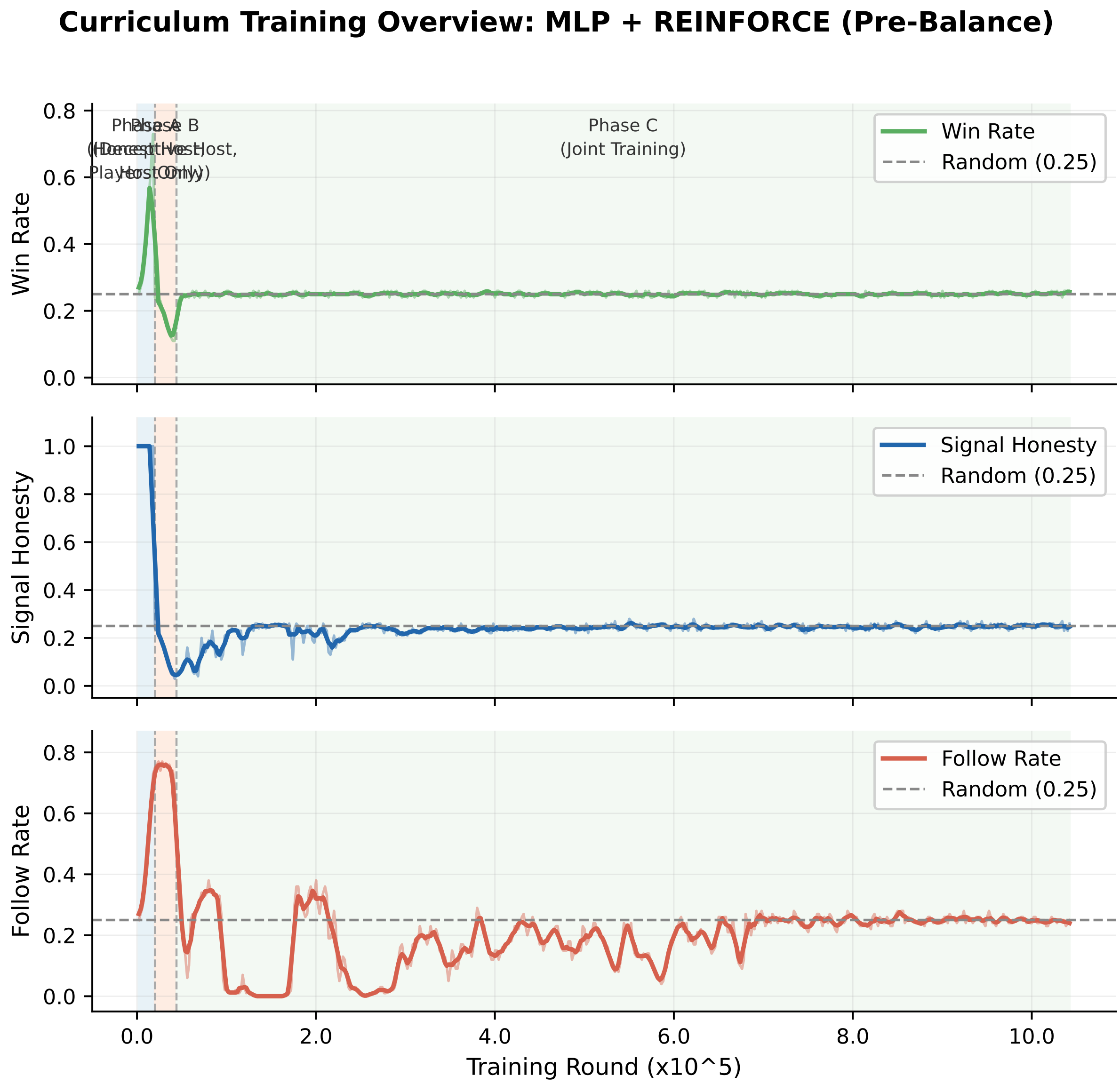


Figure 1: **Curriculum training overview (MLP+REINFORCE, pre-balance).** Win rate, signal honesty, and follow rate across all three phases. Shaded regions indicate Phase A (blue), Phase B (orange), and Phase C (green); dashed vertical lines mark phase transitions. The grey dashed line marks the random-chance baseline (0.25).

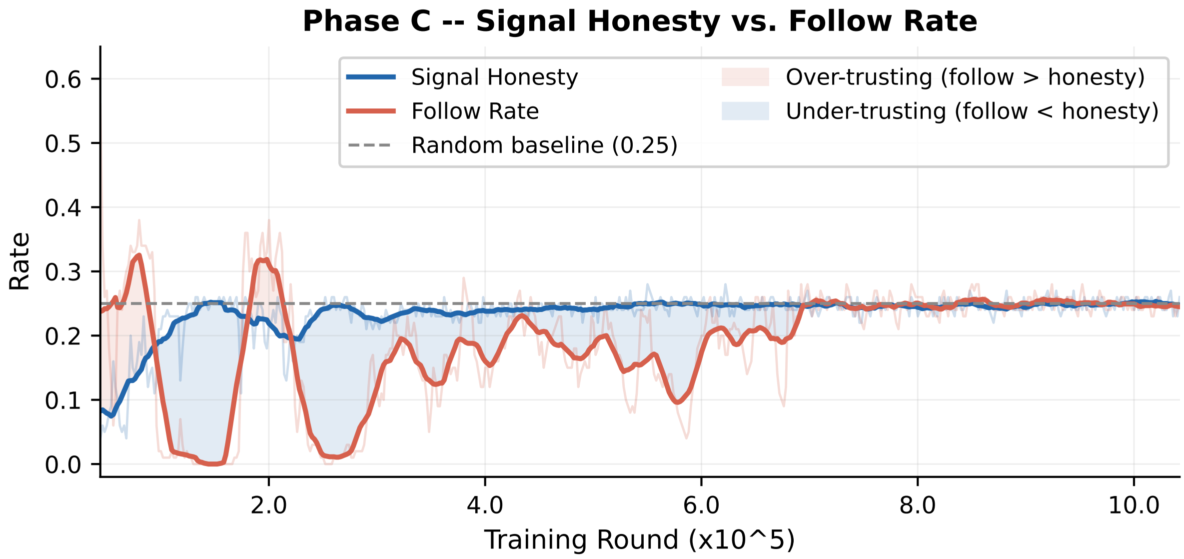


Figure 2: **Phase C – Signal Honesty vs. Follow Rate.** Thin lines show raw per-checkpoint values; thick lines show MA-15 smoothed trends. Red shading indicates periods when players over-trust the host ($follow > honesty$); blue shading indicates under-trust. Both metrics converge to the mixed Nash equilibrium at $\approx 0.25$.

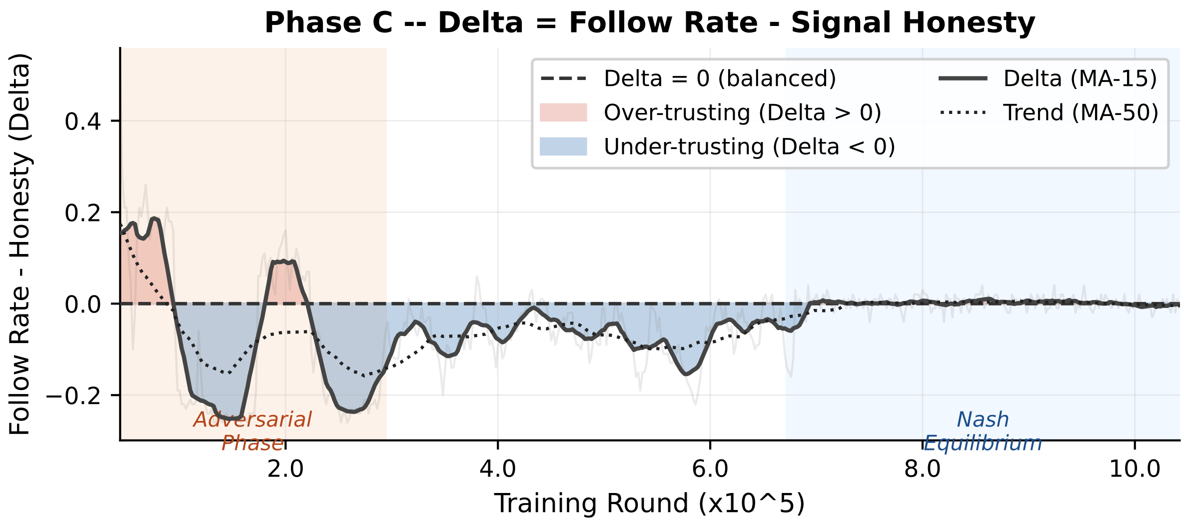


Figure 3: **Phase C –** $\Delta =$ **Follow Rate – Signal Honesty.** The dotted line shows the MA-50 trend. The orange band marks the early adversarial phase (large oscillations); the blue band marks convergence toward Nash equilibrium ($\Delta \to 0$).